

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination–2024****Course: B. Tech****Branch: Common To All Branches****Semester :II****Subject Code & Name: BTES204 Computer Programming in C****Max Marks: 60****Date: 26/12/24****Duration: 3Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculator is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q.1	Objective type questions.(Compulsory Question)		12
	Which of the following is a valid variable name in C?	CO2	1
	a. 2sum b. _sum2 c. sum-2 d. int		
2	What is the result of the expression (5>3&& 8<10) in C?	CO2	1
	a. true b. false c. 1 d. 0		
3	What is the output of the following C code? int x= 5, y = 10; x+=y--; printf("%d", x);	CO2	1
	a. 14 b. 15 c. 16 d. 10		
4	Which of the following is true about the switch statement in C?	CO3	1
	a. switch can handle both integers and floating-point values. b. break is necessary to exit a case. c. You can use continue inside a switch statement. d. default is required in every switch statement.		
5	Which of the following loop guarantees execution at least	CO3	1

	once?					
	a.for	b.while	c.goto	d.do-while		
6	What is the scope of a variable declared inside a function in C?				CO3	1
	a. Global	b.External	c.Local	d. Static		
7	How are elements of a one-dimensional array accessed in C?				CO4	1
	a. Using parentheses ()	b. Using curly braces { }	c. Using square brackets []	d. Using a dot .		
8	What will be the output of the following C code? intarr[5] = {1,2, 3}; printf("%d",arr[4]);				CO4	1
	a.0	b. 1	c.compilation error	d. garbage value		
9	What is the total number of elements in a 3D array declared as intarr[2][3][4];?				CO4	1
	a.6	b. 12	c.24	d. 30		
10	What is the correct way to access a structure's member using a pointer to the structure?				CO5	1
	a. struct_ptr.member	b. struct_ptr->member	c. *struct_ptr.member	d. struct_ptr&member		
11	How can you pass a structure to a function in C?				CO5	1
	a. Bypassing individual members of the structure.	b. Bypassing the entire structure by value.	c. Bypassing a pointer to the structure.	d. All of the above.		
12	Which of the following is correct about pointers to structures in C?				CO5	1
	a. They cannot be used to modify structure members.	b. They can be dereferenced using the-> operator.	c. They are declared using the new keyword.	d. They are invalid in C.		

Q.2	Solve the following.		12
A)	Differentiate between i) High level and Low level language. ii) Compiler and Interpreter	CO1	6
B)	Draw the flowchart, write algorithm to check the given number is odd or even. What are the advantages of flowchart and algorithm?	CO1	6
Q.3	Solve the following.		12
A)	Write a program to print area of triangle, circle, square.	CO2	6
B)	Explain Tokens with example.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	Write a C program to find factorial of a number using functions.	CO3	6
B)	Write the program using Switch Statement which takes an arithmetic operator +, -, *, / and two operands from the user as an input and performs the calculation on the two operands.	CO3	6
C)	What is function in C? Write the syntax for function prototype, function definition and function call.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Define Array. What are various types of array and write syntax of any two of them.	CO4	6
B)	Write a program to read 0 to 10 numbers from the user by using array and display average of these numbers.	CO4	6
C)	Define string in C? Write syntax to declare character array. How to initialize Character array?	CO4	6
Q.6	Solve Any Two of the following.		12
A)	Define structure in C? What are the different ways to declare structure variables?	CO5	6
B)	Differentiate between array and structure with suitable example.	CO5	6
C)	Write a C program to store information (name, Roll no and Marks) of a student and displays it on the screen using structure.	CO5	6
	End		